

Solução do Problema stair - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: stair
- Número de Variáveis: 467
- Número de Restrições: 356
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: -2.513e+02
- Valor Dual: -9.077e+01
- Gap: -7.413e-18
- Número de Iterações: 16

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema stair

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	3.040e+00	1	0.000e+00
2	5.500e+01	2	-1.651e-01
3	2.088e+02	3	-7.215e-01
4	2.602e+00	4	0.000e+00
5	2.140e-01	5	0.000e+00
6	1.123e+00	6	0.000e+00
7	3.469e+00	7	0.000e+00
8	1.523e+00	8	0.000e+00
9	6.802e-01	9	0.000e+00
10	1.690e+00	10	0.000e+00
11	5.524e-01	11	0.000e+00
12	1.249e+00	12	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	3.844e+00	13	0.000e+00
14	2.696e+00	14	0.000e+00
15	6.653e-01	15	0.000e+00
16	6.505e+00	16	0.000e+00
17	7.014e-01	17	0.000e+00
18	5.449e+00	18	0.000e+00
19	3.040e+00	19	0.000e+00
20	2.513e+02	20	0.000e+00
21	4.247e+01	21	0.000e+00
22	1.097e-03	22	0.000e+00
23	7.371e-03	23	0.000e+00
24	5.472e-02	24	0.000e+00
25	6.146e-01	25	0.000e+00
26	0.000e+00	26	4.626e+00
27	0.000e+00	27	1.682e+00
28	0.000e+00	28	1.682e+00
29	0.000e+00	29	3.842e-01
30	9.506e+00	30	0.000e+00
31	9.545e-02	31	0.000e+00
32	7.738e-01	32	0.000e+00
33	1.633e+00	33	0.000e+00
34	6.280e+00	34	0.000e+00
35	6.774e+00	35	0.000e+00
36	4.342e+00	36	3.676e-03
37	2.655e+00	37	-4.281e-02
38	4.706e-01	38	-4.425e-02
39	3.430e+00	39	-4.206e-02
40	6.619e-01	40	-3.255e-02
41	3.692e-01	41	-1.273e-02
42	8.785e-01	42	-3.135e-02
43	1.491e-01	43	-2.842e-02
44	3.061e-01	44	-2.678e-02
45	7.868e-01	45	-1.567e-02
46	1.367e+00	46	0.000e+00
47	1.781e+01	47	0.000e+00
48	5.465e+00	48	1.001e-02
49	2.500e+00	49	0.000e+00
50	4.300e+00	50	-1.268e-01
51	5.000e+00	51	-2.651e-02
52	4.900e+00	52	1.268e-01
53	7.157e+01	53	0.000e+00
54	5.210e+01	54	0.000e+00
55	2.609e+02	55	0.000e+00
56	5.258e+01	56	0.000e+00
57	5.716e+00	57	0.000e+00
58	2.037e+01	58	0.000e+00
59	5.909e+01	59	0.000e+00
60	2.724e+01	60	0.000e+00
61	1.187e+01	61	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	2.298e+01	62	0.000e+00
63	7.008e+00	63	0.000e+00
64	1.452e+01	64	0.000e+00
65	3.876e+01	65	0.000e+00
66	3.486e+01	66	0.000e+00
67	7.665e+00	67	0.000e+00
68	1.094e+02	68	0.000e+00
69	1.470e+01	69	0.000e+00
70	8.859e+01	70	0.000e+00
71	1.391e+00	71	0.000e+00
72	2.442e-01	72	0.000e+00
73	1.510e+00	73	0.000e+00
74	4.526e+00	74	0.000e+00
75	2.313e+00	75	0.000e+00
76	8.627e-01	76	0.000e+00
77	2.443e+00	77	0.000e+00
78	5.379e-01	78	0.000e+00
79	1.634e+00	79	0.000e+00
80	4.609e+00	80	0.000e+00
81	1.346e+00	81	0.000e+00
82	9.226e-01	82	0.000e+00
83	7.581e+00	83	0.000e+00
84	8.626e-01	84	0.000e+00
85	7.113e+00	85	0.000e+00
86	0.000e+00	86	0.000e+00
87	1.381e-01	87	0.000e+00
88	1.559e-02	88	0.000e+00
89	1.255e+01	89	0.000e+00
90	5.000e+00	90	0.000e+00
91	6.907e+01	91	0.000e+00
92	3.033e+02	92	0.000e+00
93	5.202e+01	93	0.000e+00
94	3.203e+02	94	0.000e+00
95	6.592e-03	95	0.000e+00
96	6.683e-02	96	0.000e+00
97	2.203e-01	97	0.000e+00
98	1.837e+00	98	0.000e+00
99	0.000e+00	99	2.746e+00
100	0.000e+00	100	9.984e-01
101	0.000e+00	101	9.984e-01
102	0.000e+00	102	2.375e-01
103	0.000e+00	103	3.745e-03
104	1.163e-01	104	0.000e+00
105	9.272e-01	105	0.000e+00
106	1.916e+00	106	0.000e+00
107	7.134e+00	107	0.000e+00
108	6.004e+00	108	0.000e+00
109	4.884e+00	109	7.800e-03
110	2.986e+00	110	-2.850e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	4.920e-01	111	-2.799e-02
112	4.085e+00	112	-2.625e-02
113	7.553e-01	113	-2.103e-02
114	4.397e-01	114	-8.396e-03
115	1.169e+00	115	-1.902e-02
116	1.852e-01	116	-1.812e-02
117	4.075e-01	117	-1.449e-02
118	1.105e+00	118	-1.022e-02
119	3.122e+00	119	0.000e+00
120	2.268e+01	120	0.000e+00
121	7.077e+00	121	6.132e-03
122	-5.000e-03	122	0.000e+00
123	4.700e+00	123	-7.396e-02
124	5.500e+00	124	-1.223e-02
125	4.900e+00	125	7.396e-02
126	8.163e+01	126	0.000e+00
127	5.675e+01	127	0.000e+00
128	6.448e+00	128	0.000e+00
129	2.490e+01	129	0.000e+00
130	7.267e+01	130	0.000e+00
131	3.418e+01	131	0.000e+00
132	1.446e+01	132	0.000e+00
133	3.031e+01	133	0.000e+00
134	8.622e+00	134	0.000e+00
135	1.942e+01	135	0.000e+00
136	5.259e+01	136	0.000e+00
137	3.890e+01	137	0.000e+00
138	1.043e+01	138	0.000e+00
139	1.321e+02	139	0.000e+00
140	1.729e+01	140	0.000e+00
141	1.099e+02	141	0.000e+00
142	3.235e+00	142	0.000e+00
143	3.383e-01	143	0.000e+00
144	2.096e+00	144	0.000e+00
145	5.933e+00	145	0.000e+00
146	3.039e+00	146	0.000e+00
147	1.224e+00	147	0.000e+00
148	3.493e+00	148	0.000e+00
149	9.695e-01	149	0.000e+00
150	2.540e+00	150	0.000e+00
151	6.822e+00	151	0.000e+00
152	3.432e+00	152	0.000e+00
153	1.322e+00	153	0.000e+00
154	9.811e+00	154	0.000e+00
155	1.056e+00	155	0.000e+00
156	7.492e+00	156	0.000e+00
157	1.559e-02	157	0.000e+00
158	8.503e-02	158	0.000e+00
159	3.247e-01	159	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	0.000e+00	160	1.330e+00
161	1.381e-01	161	0.000e+00
162	1.985e-02	162	0.000e+00
163	1.255e+01	163	0.000e+00
164	1.050e+01	164	0.000e+00
165	8.164e+01	165	0.000e+00
166	3.670e+02	166	0.000e+00
167	6.373e+01	167	0.000e+00
168	3.849e+02	168	0.000e+00
169	1.618e+00	169	0.000e+00
170	0.000e+00	170	2.141e+00
171	0.000e+00	171	1.327e+00
172	0.000e+00	172	6.027e-01
173	0.000e+00	173	1.294e-01
174	4.801e+00	174	0.000e+00
175	1.453e-01	175	0.000e+00
176	1.101e+00	176	0.000e+00
177	2.264e+00	177	0.000e+00
178	8.268e+00	178	0.000e+00
179	6.703e+00	179	0.000e+00
180	5.494e+00	180	2.640e-03
181	3.359e+00	181	-1.317e-02
182	5.145e-01	182	-1.438e-02
183	4.866e+00	183	-1.433e-02
184	8.619e-01	184	-1.106e-02
185	5.237e-01	185	-4.305e-03
186	1.556e+00	186	-9.111e-03
187	2.301e-01	187	-8.963e-03
188	5.423e-01	188	-6.607e-03
189	1.553e+00	189	-3.189e-03
190	6.482e+00	190	0.000e+00
191	3.001e+01	191	0.000e+00
192	9.165e+00	192	5.085e-03
193	-3.970e+00	193	0.000e+00
194	4.300e+00	194	-4.416e-02
195	5.500e+00	195	-5.442e-03
196	4.900e+00	196	4.416e-02
197	1.050e+02	197	0.000e+00
198	6.646e+01	198	0.000e+00
199	7.463e+00	199	0.000e+00
200	3.119e+01	200	0.000e+00
201	9.047e+01	201	0.000e+00
202	4.330e+01	202	0.000e+00
203	1.813e+01	203	0.000e+00
204	4.079e+01	204	0.000e+00
205	1.153e+01	205	0.000e+00
206	2.704e+01	206	0.000e+00
207	7.306e+01	207	0.000e+00
208	4.919e+01	208	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	1.440e+01	209	0.000e+00
210	1.615e+02	210	0.000e+00
211	2.046e+01	211	0.000e+00
212	1.324e+02	212	0.000e+00
213	3.171e+00	213	0.000e+00
214	3.673e-01	214	0.000e+00
215	2.560e+00	215	0.000e+00
216	7.006e+00	216	0.000e+00
217	3.783e+00	217	0.000e+00
218	1.422e+00	218	0.000e+00
219	4.443e+00	219	0.000e+00
220	1.118e+00	220	0.000e+00
221	3.207e+00	221	0.000e+00
222	8.782e+00	222	0.000e+00
223	3.405e+00	223	0.000e+00
224	1.782e+00	224	0.000e+00
225	1.159e+01	225	0.000e+00
226	1.233e+00	226	0.000e+00
227	9.620e+00	227	0.000e+00
228	1.985e-02	228	0.000e+00
229	6.199e-02	229	0.000e+00
230	3.573e-01	230	0.000e+00
231	1.559e-02	231	0.000e+00
232	8.503e-02	232	0.000e+00
233	4.628e-01	233	0.000e+00
234	1.925e-02	234	0.000e+00
235	1.735e+01	235	0.000e+00
236	1.089e+02	236	0.000e+00
237	1.600e+01	237	0.000e+00
238	4.451e+02	238	0.000e+00
239	7.807e+01	239	0.000e+00
240	4.760e+02	240	0.000e+00
241	1.999e+00	241	0.000e+00
242	0.000e+00	242	1.073e+00
243	0.000e+00	243	9.914e-01
244	0.000e+00	244	2.057e-01
245	0.000e+00	245	8.763e-02
246	5.158e+00	246	0.000e+00
247	1.823e-01	247	0.000e+00
248	1.311e+00	248	0.000e+00
249	2.672e+00	249	0.000e+00
250	9.505e+00	250	0.000e+00
251	6.831e+00	251	0.000e+00
252	6.180e+00	252	3.652e-03
253	3.778e+00	253	-8.792e-03
254	5.380e-01	254	-9.673e-03
255	5.795e+00	255	-9.581e-03
256	9.836e-01	256	-7.132e-03
257	6.238e-01	257	-3.021e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	2.071e+00	258	-6.789e-03
259	2.858e-01	259	-5.517e-03
260	7.218e-01	260	-4.493e-03
261	2.182e+00	261	-2.512e-03
262	9.472e+00	262	0.000e+00
263	3.769e+01	263	0.000e+00
264	1.187e+01	264	3.237e-03
265	-8.636e+00	265	0.000e+00
266	3.100e+00	266	-2.713e-02
267	5.500e+00	267	-2.547e-03
268	4.900e+00	268	2.713e-02
269	1.286e+02	269	0.000e+00
270	7.597e+01	270	0.000e+00
271	8.565e+00	271	0.000e+00
272	3.887e+01	272	0.000e+00
273	1.115e+02	273	0.000e+00
274	5.465e+01	274	0.000e+00
275	2.239e+01	275	0.000e+00
276	5.412e+01	276	0.000e+00
277	1.488e+01	277	0.000e+00
278	3.666e+01	278	0.000e+00
279	9.941e+01	279	0.000e+00
280	5.941e+01	280	0.000e+00
281	1.975e+01	281	0.000e+00
282	1.963e+02	282	0.000e+00
283	2.416e+01	283	0.000e+00
284	1.613e+02	284	0.000e+00
285	3.904e+00	285	0.000e+00
286	4.005e-01	286	0.000e+00
287	3.149e+00	287	0.000e+00
288	8.512e+00	288	0.000e+00
289	4.767e+00	289	0.000e+00
290	1.668e+00	290	0.000e+00
291	5.747e+00	291	0.000e+00
292	1.291e+00	292	0.000e+00
293	4.011e+00	293	0.000e+00
294	1.113e+01	294	0.000e+00
295	3.235e+00	295	0.000e+00
296	2.350e+00	296	0.000e+00
297	1.363e+01	297	0.000e+00
298	1.435e+00	298	0.000e+00
299	1.167e+01	299	0.000e+00
300	1.925e-02	300	0.000e+00
301	1.047e-01	301	0.000e+00
302	3.200e-01	302	0.000e+00
303	3.544e-02	303	0.000e+00
304	1.470e-01	304	0.000e+00
305	8.201e-01	305	0.000e+00
306	3.136e-02	306	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
307	2.250e+01	307	0.000e+00
308	2.150e+01	308	0.000e+00
309	1.373e+02	309	0.000e+00
310	5.407e+02	310	0.000e+00
311	9.564e+01	311	0.000e+00
312	5.824e+02	312	0.000e+00
313	2.483e+00	313	0.000e+00
314	0.000e+00	314	1.150e+00
315	0.000e+00	315	1.320e-01
316	0.000e+00	316	3.142e-01
317	0.000e+00	317	5.078e-02
318	1.133e+01	318	0.000e+00
319	2.226e-01	319	0.000e+00
320	1.558e+00	320	0.000e+00
321	3.139e+00	321	0.000e+00
322	1.084e+01	322	0.000e+00
323	6.887e+00	323	0.000e+00
324	6.952e+00	324	1.730e-03
325	4.250e+00	325	-5.183e-03
326	5.626e-01	326	-5.775e-03
327	6.902e+00	327	-5.913e-03
328	1.122e+00	328	-4.386e-03
329	7.429e-01	329	-2.150e-03
330	2.757e+00	330	-3.892e-03
331	3.551e-01	331	-3.011e-03
332	9.607e-01	332	-2.521e-03
333	3.065e+00	333	-1.175e-03
334	1.284e+01	334	0.000e+00
335	4.679e+01	335	0.000e+00
336	1.537e+01	336	1.636e-03
337	-1.550e+01	337	0.000e+00
338	1.500e+00	338	-1.616e-02
339	3.500e+00	339	0.000e+00
340	4.900e+00	340	1.616e-02
341	1.525e+02	341	0.000e+00
342	8.768e+01	342	0.000e+00
343	9.767e+00	343	0.000e+00
344	4.832e+01	344	0.000e+00
345	1.370e+02	345	0.000e+00
346	6.895e+01	346	0.000e+00
347	2.740e+01	347	0.000e+00
348	7.136e+01	348	0.000e+00
349	1.876e+01	349	0.000e+00
350	4.869e+01	350	0.000e+00
351	1.328e+02	351	0.000e+00
352	6.911e+01	352	0.000e+00
353	2.679e+01	353	0.000e+00
354	2.372e+02	354	0.000e+00
355	2.846e+01	355	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
356	1.963e+02	356	0.000e+00
357	3.086e+00	357	0.000e+00
358	4.300e-01	358	0.000e+00
359	3.693e+00	359	0.000e+00
360	8.650e+00	360	0.000e+00
361	5.629e+00	361	0.000e+00
362	1.785e+00	362	0.000e+00
363	6.541e+00	363	0.000e+00
364	1.544e+00	364	0.000e+00
365	4.683e+00	365	0.000e+00
366	1.367e+01	366	0.000e+00
367	4.076e+00	367	0.000e+00
368	2.828e+00	368	0.000e+00
369	1.538e+01	369	0.000e+00
370	1.661e+00	370	0.000e+00
371	1.344e+01	371	0.000e+00
372	3.136e-02	372	0.000e+00
373	9.024e-02	373	0.000e+00
374	3.986e-01	374	0.000e+00
375	5.469e-02	375	0.000e+00
376	2.517e-01	376	0.000e+00
377	1.140e+00	377	0.000e+00
378	1.720e-02	378	0.000e+00
379	3.383e+01	379	0.000e+00
380	2.500e+01	380	0.000e+00
381	1.680e+02	381	0.000e+00
382	6.579e+02	382	0.000e+00
383	1.172e+02	383	0.000e+00
384	7.087e+02	384	0.000e+00
385	2.477e+00	385	0.000e+00
386	6.185e-03	386	0.000e+00
387	0.000e+00	387	5.577e-01
388	0.000e+00	388	4.709e-02
389	0.000e+00	389	8.606e-02
390	0.000e+00	390	1.151e-02
391	2.720e-01	391	0.000e+00
392	1.831e+00	392	0.000e+00
393	3.645e+00	393	0.000e+00
394	1.223e+01	394	0.000e+00
395	6.604e+00	395	0.000e+00
396	7.820e+00	396	1.559e-03
397	4.781e+00	397	-7.530e-03
398	5.883e-01	398	-8.765e-03
399	8.220e+00	399	-9.624e-03
400	1.281e+00	400	-6.868e-03
401	8.848e-01	401	-3.263e-03
402	3.669e+00	402	-6.298e-03
403	4.411e-01	403	-3.861e-03
404	1.279e+00	404	-3.434e-03

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
405	4.306e+00	405	-1.430e-03
406	9.026e+00	406	0.000e+00
407	4.885e+01	407	0.000e+00
408	1.991e+01	408	3.794e-03
409	-1.762e+01	409	0.000e+00
410	-5.000e-01	410	-2.554e-02
411	5.000e+00	411	-2.268e-02
412	4.900e+00	412	2.554e-02
413	1.751e+02	413	0.000e+00
414	1.928e+02	414	0.000e+00
415	1.435e+02	415	0.000e+00
416	8.507e+02	416	0.000e+00
417	9.694e+01	417	0.000e+00
418	1.106e+01	418	0.000e+00
419	5.940e+01	419	0.000e+00
420	1.630e+02	420	0.000e+00
421	8.584e+01	421	0.000e+00
422	3.275e+01	422	0.000e+00
423	9.099e+01	423	0.000e+00
424	2.339e+01	424	0.000e+00
425	6.274e+01	425	0.000e+00
426	1.738e+02	426	0.000e+00
427	8.134e+01	427	0.000e+00
428	3.528e+01	428	0.000e+00
429	2.834e+02	429	0.000e+00
430	3.345e+01	430	0.000e+00
431	2.366e+02	431	0.000e+00
432	7.103e+00	432	0.000e+00
433	6.443e-01	433	0.000e+00
434	4.256e+00	434	0.000e+00
435	1.134e+01	435	0.000e+00
436	6.127e+00	436	0.000e+00
437	2.324e+00	437	0.000e+00
438	6.589e+00	438	0.000e+00
439	1.707e+00	439	0.000e+00
440	4.620e+00	440	0.000e+00
441	1.287e+01	441	0.000e+00
442	6.083e+00	442	0.000e+00
443	2.550e+00	443	0.000e+00
444	2.057e+01	444	0.000e+00
445	2.430e+00	445	0.000e+00
446	1.670e+01	446	0.000e+00
447	1.720e-02	447	0.000e+00
448	1.236e-01	448	0.000e+00
449	2.293e-01	449	0.000e+00
450	8.605e-02	450	0.000e+00
451	3.420e-01	451	0.000e+00
452	1.539e+00	452	0.000e+00
453	3.383e+01	453	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
454	8.014e+02	454	0.000e+00
455	1.033e-01	455	0.000e+00
456	4.656e-01	456	0.000e+00
457	1.768e+00	457	0.000e+00
458	2.124e+00	458	0.000e+00
459	0.000e+00	459	9.725e-01
460	0.000e+00	460	3.912e-01
461	0.000e+00	461	2.880e-01
462	0.000e+00	462	0.000e+00
463	3.203e-01	463	0.000e+00
464	2.158e+00	464	0.000e+00
465	4.237e+00	465	0.000e+00
466	1.377e+01	466	0.000e+00
467	8.428e+00	467	0.000e+00

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema stair

Coordenada y	Valor y (Dual)
1	-1.651e-01
2	1.651e-01
3	-7.215e-01
4	-1.392e-01
5	-9.271e-02
6	-9.126e-02
7	-9.345e-02
8	-1.030e-01
9	-1.228e-01
10	-1.042e-01
11	-1.071e-01
12	-1.087e-01
13	-1.198e-01
14	-9.672e-02
15	-1.525e-01
16	-5.314e-02
17	-2.791e-01
18	-1.601e-01
19	-1.234e-01
20	-5.613e-02
21	-9.526e-02
22	-5.063e-02
23	-4.430e-02
24	-7.556e-02
25	-7.077e-02
26	-7.341e-02

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Coordenada y	Valor y (Dual)
27	-9.553e-02
28	-5.393e-02
29	-1.167e-02
30	-2.495e-01
31	-2.718e-02
32	-5.151e-01
33	-1.656e-01
34	-8.832e+00
35	-4.206e+00
36	-2.523e+00
37	-8.411e-01
38	-4.569e-01
39	-8.832e+00
40	-4.206e+00
41	-2.523e+00
42	-8.411e-01
43	-4.569e-01
44	-1.355e-01
45	-1.268e-01
46	-1.268e-01
47	0.000e+00
48	0.000e+00
49	0.000e+00
50	5.890e-01
51	0.000e+00
52	9.159e-02
53	-1.003e-01
54	1.670e-01
55	-9.205e-02
56	-5.576e-02
57	-5.627e-02
58	-5.801e-02
59	-6.322e-02
60	-7.586e-02
61	-6.523e-02
62	-6.614e-02
63	-6.976e-02
64	-7.404e-02
65	-5.950e-02
66	-1.021e-01
67	-3.197e-02
68	-1.838e-01
69	-9.895e-02
70	-8.430e-02
71	-3.293e-02
72	-5.678e-02
73	-3.013e-02
74	-2.667e-02
75	-4.642e-02

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Coordenada y	Valor y (Dual)
76	-4.275e-02
77	-4.418e-02
78	-5.880e-02
79	-3.225e-02
80	-7.039e-03
81	-1.585e-01
82	-1.626e-02
83	-3.179e-01
84	-1.003e-01
85	-5.242e+00
86	-2.496e+00
87	-1.498e+00
88	-4.992e-01
89	-2.617e-01
90	-5.242e+00
91	-2.496e+00
92	-1.498e+00
93	-4.992e-01
94	-2.617e-01
95	-8.425e-02
96	-7.869e-02
97	-7.396e-02
98	4.728e-03
99	4.728e-03
100	-2.026e-03
101	3.445e-01
102	0.000e+00
103	5.193e-02
104	5.728e+00
105	2.456e+00
106	6.927e+00
107	-6.173e-02
108	9.539e-02
109	-4.479e-02
110	-2.898e-02
111	-2.776e-02
112	-2.782e-02
113	-3.108e-02
114	-3.784e-02
115	-3.304e-02
116	-3.319e-02
117	-3.554e-02
118	-3.896e-02
119	-3.144e-02
120	-5.294e-02
121	-1.576e-02
122	-8.869e-02
123	-5.392e-02
124	-4.872e-02

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Coordenada y	Valor y (Dual)
125	-1.879e-02
126	-3.238e-02
127	-1.717e-02
128	-1.502e-02
129	-2.645e-02
130	-2.405e-02
131	-2.544e-02
132	-3.370e-02
133	-1.807e-02
134	-4.103e-03
135	-9.422e-02
136	-9.584e-03
137	-1.826e-01
138	-5.716e-02
139	-4.309e+00
140	-2.167e+00
141	-8.400e-01
142	-2.373e-01
143	-1.079e-01
144	-4.309e+00
145	-2.167e+00
146	-8.400e-01
147	-2.373e-01
148	-1.079e-01
149	-4.215e-02
150	-3.943e-02
151	-4.416e-02
152	-4.728e-03
153	-4.728e-03
154	2.026e-03
155	2.033e-01
156	-6.755e-03
157	3.651e-02
158	9.757e+00
159	3.400e+00
160	1.458e+00
161	4.111e+00
162	-3.872e-02
163	6.205e-02
164	-3.267e-02
165	-2.023e-02
166	-1.934e-02
167	-1.944e-02
168	-2.189e-02
169	-2.600e-02
170	-2.223e-02
171	-2.350e-02
172	-2.452e-02
173	-2.651e-02

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Coordenada y	Valor y (Dual)
174	-2.210e-02
175	-3.671e-02
176	-1.081e-02
177	-6.249e-02
178	-3.651e-02
179	-2.876e-02
180	-1.241e-02
181	-2.113e-02
182	-1.115e-02
183	-9.524e-03
184	-1.693e-02
185	-1.517e-02
186	-1.672e-02
187	-2.137e-02
188	-1.143e-02
189	-2.730e-03
190	-6.217e-02
191	-6.535e-03
192	-1.189e-01
193	-3.759e-02
194	-2.475e+00
195	-1.401e+00
196	-4.100e-01
197	-2.043e-01
198	-1.167e-01
199	-2.475e+00
200	-1.401e+00
201	-4.100e-01
202	-2.043e-01
203	-1.167e-01
204	-2.902e-02
205	-2.713e-02
206	-2.713e-02
207	0.000e+00
208	0.000e+00
209	0.000e+00
210	1.153e-01
211	0.000e+00
212	2.081e-02
213	5.890e+00
214	2.072e+00
215	8.550e-01
216	2.607e+00
217	-2.458e-02
218	3.766e-02
219	-1.900e-02
220	-1.208e-02
221	-1.149e-02
222	-1.135e-02

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Coordenada y	Valor y (Dual)
223	-1.288e-02
224	-1.512e-02
225	-1.337e-02
226	-1.426e-02
227	-1.475e-02
228	-1.609e-02
229	-1.354e-02
230	-2.315e-02
231	-6.709e-03
232	-3.848e-02
233	-2.116e-02
234	-1.761e-02
235	-7.578e-03
236	-1.286e-02
237	-6.771e-03
238	-5.715e-03
239	-1.026e-02
240	-9.047e-03
241	-1.028e-02
242	-1.292e-02
243	-6.823e-03
244	-1.687e-03
245	-3.911e-02
246	-4.091e-03
247	-7.292e-02
248	-2.300e-02
249	-1.711e+00
250	-5.603e-01
251	-4.283e-01
252	-1.142e-01
253	-6.337e-02
254	-1.711e+00
255	-5.603e-01
256	-4.283e-01
257	-1.142e-01
258	-6.337e-02
259	-1.727e-02
260	-1.616e-02
261	-1.616e-02
262	0.000e+00
263	0.000e+00
264	0.000e+00
265	6.340e-02
266	0.000e+00
267	1.277e-02
268	3.528e+00
269	1.081e+00
270	6.493e-01
271	1.367e+00
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Coordenada y	Valor y (Dual)
272	-1.616e-02
273	2.324e-02
274	-2.858e-03
275	-2.878e-02
276	-1.969e-02
277	-1.846e-02
278	-1.760e-02
279	-2.035e-02
280	-2.396e-02
281	-2.092e-02
282	-2.336e-02
283	-2.379e-02
284	-2.579e-02
285	-2.231e-02
286	-3.907e-02
287	-1.058e-02
288	-6.131e-02
289	-3.586e-02
290	-1.026e-02
291	-4.719e-03
292	-7.968e-03
293	-4.181e-03
294	-3.482e-03
295	-6.289e-03
296	-5.494e-03
297	-6.419e-03
298	-7.891e-03
299	-4.143e-03
300	-1.070e-03
301	-2.493e-02
302	-2.619e-03
303	-4.509e-02
304	-1.410e-02
305	-6.909e-01
306	-6.909e-01
307	-1.332e-01
308	-8.606e-02
309	0.000e+00
310	-6.909e-01
311	-6.909e-01
312	-1.332e-01
313	-8.606e-02
314	0.000e+00
315	-2.722e-02
316	-2.554e-02
317	-2.554e-02
318	0.000e+00
319	0.000e+00
320	0.000e+00

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Coordenada y	Valor y (Dual)
321	3.279e-02
322	0.000e+00
323	1.277e-02
324	2.206e+00
325	9.489e-01
326	3.351e-01
327	1.198e+00
328	3.279e-02
329	-4.006e-02
330	-2.718e-02
331	-2.976e-02
332	-2.670e-02
333	-2.663e-02
334	-3.238e-02
335	-2.787e-02
336	-3.505e-02
337	-3.760e-02
338	-3.124e-02
339	-2.636e-02
340	-7.254e-02
341	-1.555e-02
342	-1.101e-01
343	-4.791e-02
344	-1.765e+00
345	-7.926e-01
346	-4.014e-01
347	-1.134e-01
348	-1.134e-01
349	-1.765e+00
350	-7.926e-01
351	-4.014e-01
352	-1.134e-01
353	-1.134e-01
354	1.481e+00
355	3.912e-01
356	2.880e-01